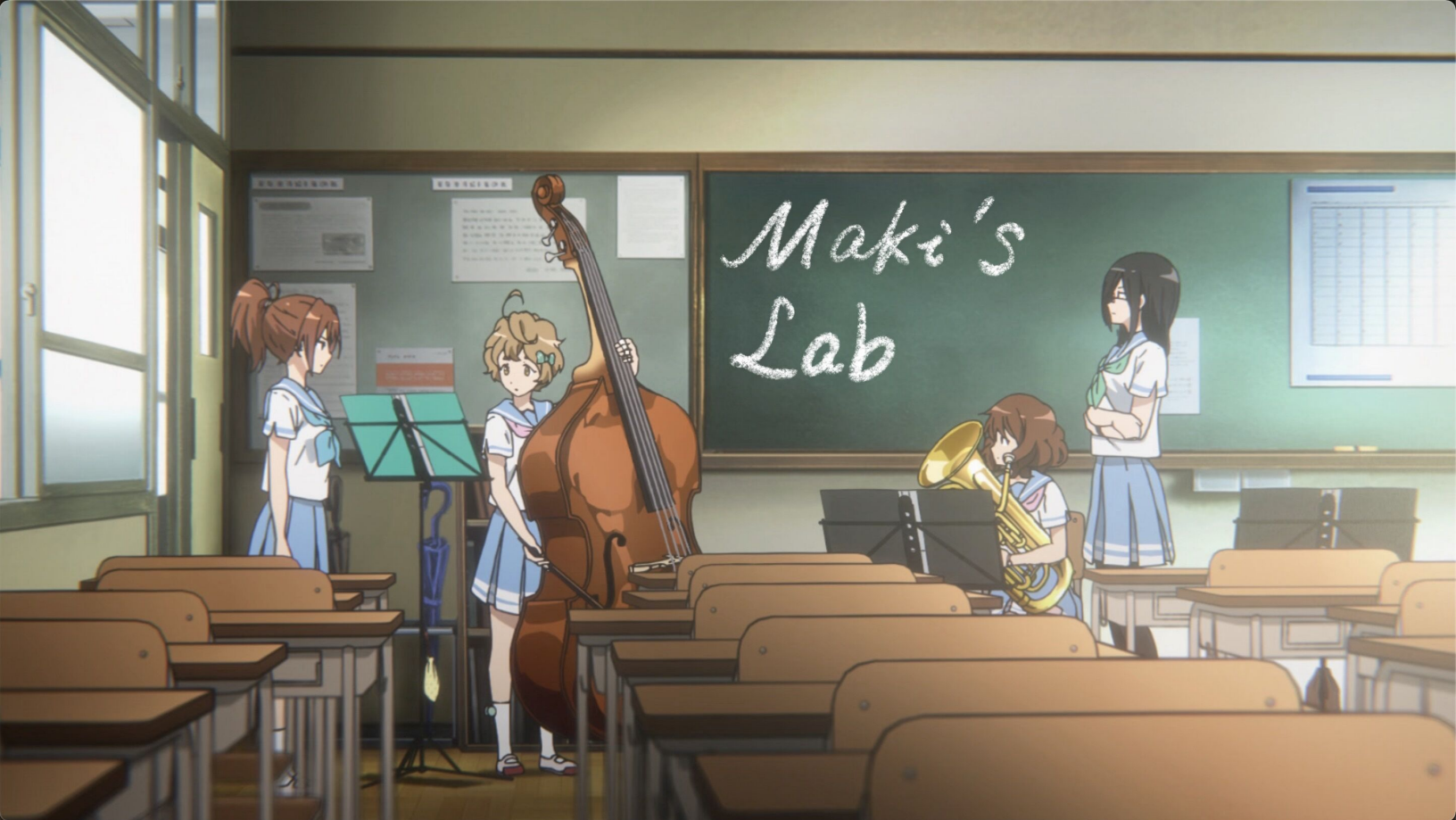


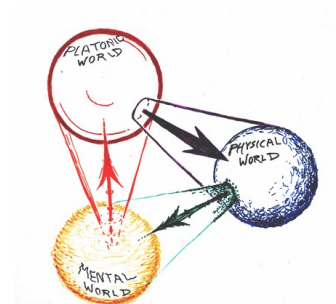


Elementary Number Theory Tutorials

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Chapter 1 Division

1.1 Notations

1. In this material, the set of natural numbers, $\mathbb{N}$, is defined to be

$$\mathbb{N} = \{1, 2, \dots\}.$$

To avoid ambiguities, we also define, for each $n \in \mathbb{Z}$, that

$$\mathbb{N}_n = \{n, n+1, \cdot\}.$$

In particular, we have $\mathbb{N} = \mathbb{N}_1$.

2. When we write $A \subset B$, what we mean is that $A \subseteq B$, or that every element in A is also in B . We use the symbol $\subsetneq$ to denote proper inclusion. In other words, $A \subsetneq B$ if and only if $A \subset B$ and $A \neq B$.

1.2 Well-Ordering Principle and Induction Principle

Axiom 1.1 (Well-Ordering Principle)

If $S \subset \mathbb{N}$ is a nonempty subset of $\mathbb{N}$, then S has a minimum. In other words, there is an element $a \in S$ such that for all $s \in S$, we have $a \leq s$.



Axiom 1.2 (Induction Principle)

Assume that for each $n \in \mathbb{N}$, we have a statement $p(n)$. Then the following are equivalent.

1. For every $n \in \mathbb{N}$, $p(n)$ is true.
2. (a). $p(1)$ is true.
(b). For every $n \in \mathbb{N}$, if $p(n)$ is true, then $p(n+1)$ is true.



Indeed, it can be shown that these two principles are equivalent. Also, both of them are equivalent to the so-called Strong Induction Principle.

Axiom 1.3 (Strong Induction Principle)

Assume that for each $n \in \mathbb{N}$, we have a statement $p(n)$. Then the following are equivalent.

1. For every $n \in \mathbb{N}$, $p(n)$ is true.
2. (a). $p(1)$ is true.
(b). For every $n \in \mathbb{N}$, if all of $p(1), \dots, p(n)$ are true, then $p(n+1)$ is true.



These axioms are fundamental for constructing some number in many proofs in this course.

We can prove the existence of Euclidean division by the Well-Ordering Principle. The uniqueness follows quickly.

Proposition 1.1 (Euclidean Division)

If $a \in \mathbb{Z}$, $b \in \mathbb{Z} - \{0\}$, then there exists unique $q, r \in \mathbb{Z}$, such that

1. $a = qb + r$, and
2. $0 \leq r \leq |b| - 1$.



1.3 Division

Definition 1.1

Assume $a \in \mathbb{Z}$, $b \in \mathbb{Z} - \{0\}$. We say that b divides a , denoted by $b|a$, if and only if there is some $c \in \mathbb{Z}$, such that $a = bc$.

Clearly, we have the following lemma.

Lemma 1.1

Assume $a \in \mathbb{Z}$, $b \in \mathbb{Z} - \{0\}$, then $b|a$ if and only if the remainder of the corresponding Euclidean division is 0.

Also, we have a lemma to describe the relation of division.

Lemma 1.2

If we restrict the division to natural numbers, then the relation of division is a partial order on $\mathbb{N}$. To be more precise, for any $a, b, c \in \mathbb{N}$, we have the following properties.

1. $a|a$.
2. If $a|b$ and $b|a$, then $a = b$.
3. If $a|b$ and $b|c$, then $a|c$.

We have some other useful properties of division.

Lemma 1.3

Assume $a, b, c, d \in \mathbb{Z} - \{0\}$, $k, l \in \mathbb{Z}$, $n \in \mathbb{N}$.

1. $a|0$.
2. If $a|b$ and $b \neq 0$, then $1 \leq |a| \leq |b|$.
3. If $a|b$ and $a|c$, then $a|(kb + lc)$.
4. If $a|b$ and $c|d$, then $ac|bd$.
5. If $a|b$, then $a^n|b^n$.
6. If $a|bc$ and $b|a$, then $(a/b)|c$.

1.4 Greatest Common Divisor and Least Common Multiple

Definition 1.2

Assume $a, b \in \mathbb{Z} - \{0\}$.

1. The greatest common divisor of a, b , denoted by $\gcd(a, b)$, is defined to be the largest positive integer d such that $d|a$ and $d|b$.
2. The least common multiple of a, b , denoted by $\text{lcm}(a, b)$, is defined to be the smallest positive integer m such that $a|m$ and $b|m$.

Clearly, by definition, we have the following.

Lemma 1.4

Assume $a, b \in \mathbb{Z} - \{0\}$, $e, n \in \mathbb{N}$.

1. If $e|a$ and $e|b$, then $e \leq d$.
2. If $a|n$ and $b|n$, then $m \leq n$.

In the future, we can prove a stronger proposition.

Proposition 1.2

Assume $a, b \in \mathbb{Z} - \{0\}$, $e, n \in \mathbb{N}$.

1. If $e|a$ and $e|b$, then $e|d$.
2. If $a|n$ and $b|n$, then $m|n$.

Using this proposition, we can prove the following.

Proposition 1.3

Assume $a, b \in \mathbb{N}$, then

$$\gcd(a, b) \operatorname{lcm}(a, b) = ab.$$

1.5 Euclidean Algorithm

Proposition 1.4

Assume $a = r_0 \in \mathbb{Z}$, $b = r_1 \in \mathbb{Z} - \{0\}$, then we may apply the Euclidean algorithm to get

$$r_0 = q_1 r_1 + r_2$$

$$r_1 = q_2 r_2 + r_3$$

...

$$r_{n-2} = q_{n-1} r_{n-1} + r_n$$

$$r_{n-1} = q_n r_n$$

where all q_i 's and r_i 's are integers, and $0 < r_n < \dots < r_2 < |r_1|$.

It can be shown that the $r_n = \gcd(a, b)$. To show this, we need a simple lemma.

Lemma 1.5

Assume $a, b \in \mathbb{Z} - \{0\}$, $q \in \mathbb{Z}$, then

$$\gcd(a, b) = \gcd(b, a - qb).$$

Now, we can easily prove the result by induction.

Proposition 1.5

Under the assumptions of the previous proposition, we have

$$r_n = \gcd(a, b).$$

Also, as a result, this gives Bezout's identity.

Proposition 1.6

Assume $a, b \in \mathbb{Z} - \{0\}$, then there exists $m, n \in \mathbb{Z}$, such that

$$ma + nb = \gcd(a, b).$$

In particular, if $\gcd(a, b) = 1$, then we can find $m, n \in \mathbb{Z}$ such that $ma + nb = 1$.

Definition 1.3

Assume $a, b \in \mathbb{Z} - \{0\}$. We say that a and b are coprime if and only if $\gcd(a, b) = 1$. On the other hand, this means that the only positive common divisor of a and b is 1.

Lemma 1.6

If $a, b \in \mathbb{Z} - \{0\}$ are coprime, then for every $c \in \mathbb{Z}$, we can find $m, n \in \mathbb{Z}$, such that

$$ma + nb = c.$$

In general, we can show the following result.

Proposition 1.7

Assume $a, b \in \mathbb{Z} - \{0\}$ and $c \in \mathbb{Z}$, then the following are equivalent.

1. $ma + nb = c$ has integer solutions $m, n \in \mathbb{Z}$.
2. $\gcd(a, b) | c$.

Both sides are not hard to prove.

Now, we may prove some properties of GCD.

Proposition 1.8

Assume $a, b \in \mathbb{Z} - \{0\}$, $e, n \in \mathbb{N}$.

1. If $e|a$ and $e|b$, then $e|d$.

Now we are able to prove lots of interesting and intuitive properties.

Proposition 1.9

Assume $a, b, c \in \mathbb{Z} - \{0\}$.

1. $\gcd(ca, cb) = c \gcd(a, b)$.
2. If $c|a$ and $c|b$, then

$$\gcd\left(\frac{a}{c}, \frac{b}{c}\right) = \frac{\gcd(a, b)}{c}.$$

In particular,

$$\gcd\left(\frac{a}{\gcd(a, b)}, \frac{b}{\gcd(a, b)}\right) = 1.$$

3. If $a|bc$ and $\gcd(a, b) = 1$, then $a|c$.
4. If $a|bc$, then

$$\frac{a}{\gcd(a, b)} \mid c.$$

5. $\gcd(\gcd(a, b), c) = \gcd(a, \gcd(b, c))$.

Likewise, we can easily generalize the well-behaved notion of greatest common divisor to any finitely many nonzero integers.

1.6 *Subgroups of $\mathbb{Z}$

We may study the GCD and LCM using some group theory. This will make it a lot easier to prove some properties.

Definition 1.4

$G \subset \mathbb{Z}$ is called a subgroup of $\mathbb{Z}$ if and only if

1. $0 \in G$.
2. If $a, b \in G$, then $a - b \in G$.

Clearly, we have the following lemma.

Lemma 1.7

If G is a subgroup of $\mathbb{Z}$, then for every $a, b \in G$, we have $-a, a + b \in G$.

It can be shown immediately that every subgroup of $\mathbb{Z}$ has the form of

$$n\mathbb{Z} = \{kn : k \in \mathbb{Z}\}.$$

Definition 1.5

Assume $n \in \mathbb{Z}$, then $n\mathbb{Z}$ is given by

$$n\mathbb{Z} = \{kn : k \in \mathbb{Z}\}.$$

Proposition 1.10

If G is a subgroup of $\mathbb{Z}$, then there exists $n \in \mathbb{N}_0$, such that $G = n\mathbb{Z}$, where n is the smallest positive integer in G , i.e.,

$$n = \min(G \cap \mathbb{N}).$$

Moreover, n divides every element in G .

Now, using Bezout's identity and the definition of least common multiple, it is readily checked the following.

Lemma 1.8

1. If $a, b \in \mathbb{Z}$, then

$$\{ma + nb : m, n \in \mathbb{Z}\}$$

is a subgroup of $\mathbb{Z}$. The smallest positive integer in this set is $\gcd(a, b)$.

2. Assume $a, b \in \mathbb{Z} - \{0\}$, then

$$\{m \in \mathbb{Z} : a|m, b|m\}$$

is a subgroup of $\mathbb{Z}$. The smallest positive integer in this set is $\text{lcm}(a, b)$.

Now, it is trivial that

1. $e|a, e|b \iff e|\gcd(a, b)$,
2. $a|m, b|m \iff \text{lcm}(a, b)|m$.

The following proposition can be easily proved now.

Proposition 1.11

Assume $a, b \in \mathbb{N}$, then

$$\gcd(a, b) \text{lcm}(a, b) = ab.$$

Chapter 2 Prime Numbers

2.1 Prime Numbers and Composite Numbers

Definition 2.1

Assume $p \in \mathbb{N}_2$. We say that p is a prime number or p is prime, if and only if the only positive divisors of p is 1 and p .

Remark On the other hand, this means that if $p = ab$ where $a, b \in \mathbb{N}_2$, then either $a = 1$ or $b = 1$.

Definition 2.2

Assume $n \in \mathbb{N}_2$. We say that n is a composite number if and only if there exists $a, b \in \mathbb{N}_2$, such that $n = ab$.

Clearly, a positive integer is either a prime number, a composite number, or 1.

Let us examine some properties of prime numbers.

Proposition 2.1

1. If p is a prime and $a \in \mathbb{Z} - \{0\}$ such that $p \nmid a$, then $\gcd(a, p) = 1$.
2. Assume p is prime and $a, b \in \mathbb{Z}$. If $p \mid ab$, then $p \mid a$ or $p \mid b$.

We have the important theorem, called the Fundamental Theorem of Arithmetic(FTA). First, we need a lemma.

Lemma 2.1

If $n \in \mathbb{N}_2$, then n has a prime divisor.

Now we can use the properties of division and prime numbers to show the FTA.

Proposition 2.2 (FTA)

If $n \geq 2$, there exists unique prime numbers $p_1 < \dots < p_k$ and positive integers $\alpha_1, \dots, \alpha_k \in \mathbb{N}$, such that

$$n = \prod_{i=1}^k p_i^{\alpha_i} = p_1^{\alpha_1} \cdots p_k^{\alpha_k}.$$

Using FTA, we have applications to divisions and we may reprove many properties of divisions.

To simplify the procedure, we introduce the p -adic valuation notation, v_p .

Definition 2.3

Assume $n \in \mathbb{N}_2$ and

$$n = \prod_{i=1}^k p_i^{\alpha_i} = p_1^{\alpha_1} \cdots p_k^{\alpha_k}.$$

Assume p is prime. Then we define

$$v_p(n) = \begin{cases} \alpha_i, & \text{if } p = p_i \\ 0, & \text{else.} \end{cases}$$

Moreover, if $n = 1$, then it is natural to define $v_p(1) = 0$ for every prime p .

Equivalently, we may define $v_p(n)$ to be the largest $\alpha \in \mathbb{N}_0$ such that $p_\alpha \mid n$, i.e.

$$\begin{aligned} p^\alpha &\mid n \\ p^{\alpha+1} &\nmid n \end{aligned}$$

Now, using the FTA, it is clear that we have the following.

Lemma 2.2

1. If $m, n \in \mathbb{N}$, then $v_p(mn) = v_p(m) + v_p(n)$.
2. If $m, n \in \mathbb{N}$, then $m|n$ if and only if for every prime p , we have $v_p(m) \leq v_p(n)$.
3. If $m, n \in \mathbb{N}$, then $m = n$ if and only if for every prime p , we have $v_p(m) = v_p(n)$.
4. If $m, n \in \mathbb{N}$, then for every prime p , we have $v_p(\gcd(m, n)) = \min(v_p(m), v_p(n))$.
5. If $m, n \in \mathbb{N}$, then for every prime p , we have $v_p(\text{lcm}(m, n)) = \max(v_p(m), v_p(n))$.



In particular, since for any two numbers $a, b \in \mathbb{Z}$, we have that $a + b = \min(a, b) + \max(a, b)$, it is trivially true that

$$\gcd(m, n) \text{lcm}(m, n) = mn.$$

Let us reprove the following proposition.

Proposition 2.3

Assume $a, b, c \in \mathbb{Z} - \{0\}$.

1. $\gcd(ca, cb) = c \gcd(a, b)$.
2. If $c|a$ and $c|b$, then

$$\gcd\left(\frac{a}{c}, \frac{b}{c}\right) = \frac{\gcd(a, b)}{c}.$$

In particular,

$$\gcd\left(\frac{a}{\gcd(a, b)}, \frac{b}{\gcd(a, b)}\right) = 1.$$

3. If $a|bc$ and $\gcd(a, b) = 1$, then $a|c$.
4. If $a|bc$, then

$$\frac{a}{\gcd(a, b)} \Big| c.$$

5. $\gcd(\gcd(a, b), c) = \gcd(a, \gcd(b, c))$.

Likewise, we can easily generalize the well-behaved notion of greatest common divisor to any finitely many nonzero integers.



2.2 Some Estimates of Primes

To check if a number is a prime number, we need the following easy but powerful lemma.

Lemma 2.3

If $n \in \mathbb{N}_2$ is a composite number, then the smallest prime divisor of n , say p_1 , satisfies that

$$p_1 \leq \sqrt{n}.$$



Now, given a number $n \in \mathbb{N}_2$, we might be interests in the number of prime divisors of n .

Definition 2.4

Assume $n \in \mathbb{N}_2$. We define $\omega(n)$ to be the number of distinct prime divisors of n , i.e.,

$$\omega(n) = \#\{p \text{ prime} : p|n\}.$$



Remark In other words, if

$$n = \prod_{i=1}^k p_i^{\alpha_i} = p_1^{\alpha_1} \cdots p_k^{\alpha_k},$$

then $\omega(n) = k$.

Now, we have a simple upper bound of ω .

Lemma 2.4

If $n \in \mathbb{N}_2$, then $\omega(n) \leq \log_2(n)$.



2.3 Perfect Squares and Perfect Powers

First, we give the definitions of perfect squares and perfect powers. For simplicity, we restrict our attention to the positive integers.

Definition 2.5

Assume $n \in \mathbb{N}$. We say that n is a perfect square if and only if there exists some $m \in \mathbb{N}$ such that $n = m^2$.



Clearly, by the FTA, we have the following.

Lemma 2.5

Assume $n \in \mathbb{N}$, then n is a perfect square if and only if for every prime p , $v_p(n)$ is even.



Even better, we can decompose every positive integer into a product of a perfect square and a squarefree number.

Definition 2.6

Let $n \in \mathbb{N}$. We say that n is squarefree if and only if it has no divisors of the form p^2 where p is a prime. In other words, n is called squarefree if and only if for every prime p , $0 \leq v_p(n) \leq 1$.



Remark To be more precise, every squarefree positive integer n has the form $n = p_1 \cdots p_k$ where $p_1 < \cdots < p_k$ are distinct primes.

It is now not hard to show the property of decomposition.

Proposition 2.4

Assume $n \in \mathbb{N}$, then there exists unique $m, q \in \mathbb{N}$, such that

1. $n = m^2 q$.
2. q is squarefree, or there exists distinct primes $p_1 < \cdots < p_k$, such that $q = p_1 \cdots p_k$.



Similarly, we have the analogous notions for any perfect powers.

Definition 2.7

Assume $n \in \mathbb{N}$. We say that n is a perfect k -th power if and only if there exists some $m \in \mathbb{N}$ such that $n = m^k$.

**Lemma 2.6**

Assume $n \in \mathbb{N}$, then n is a perfect k -th power if and only if for every prime p , $k | v_p(n)$.

**Definition 2.8**

Let $n \in \mathbb{N}$. We say that n is k -th power free if and only if it has no divisors of the form p^k where p is a prime. In other words, n is called k -th power free if and only if for every prime p , $0 \leq v_p(n) \leq k$.



Remark To be more precise, every k -th power free positive integer n has the form $n = p_1^{\alpha_1} \cdots p_l^{\alpha_l}$ where $p_1 < \cdots < p_l$ are distinct primes and for each $1 \leq i \leq l$, we have that $1 \leq \alpha_i \leq k$.

It is similar to show the corresponding decomposition.

Proposition 2.5

Assume $n \in \mathbb{N}$, then there exists unique $m, q \in \mathbb{N}$, such that

1. $n = m^k q$.
2. q is k -th power free, or there exists distinct primes $p_1 < \dots < p_l$, and $1 \leq \alpha_1, \dots, \alpha_l \leq k$, such that

$$n = p_1^{\alpha_1} \dots p_l^{\alpha_l}.$$

2.4 Linear Congruence Equation

Now consider the equation $ax + by = c$. We want to find all solutions to this equation for any given $a, b \in \mathbb{Z} - \{0\}$ and $c \in \mathbb{Z}$.

Without loss of generality, let us assume that $a, b, c \in \mathbb{N}$.

By Bezout's Identity we know that this equation has a solution if and only if $\gcd(a, b) | c$.

Now, let us assume that $\gcd(a, b) | c$. We may now divide both sides by $\gcd(a, b)$, which means that we can further assume that $\gcd(a, b) = 1$.

1. By Bezout's Identity or Euclidean Algorithm, we can find $x_0, y_0 \in \mathbb{Z}$ such that $ax_0 + by_0 = \gcd(a, b) = 1$. Therefore, by multiplying an obvious factor, we get a particular solution to $ax + by = c$. In precise, let $x_1 = cx_0$ and $y_1 = cy_0$, then $ax_1 + by_1 = c$.
2. Now assume $ax + by = c$ for some $x, y \in \mathbb{Z}$. Since $ax_1 + by_1 = c$, we have that

$$a(x - x_1) = b(y_1 - y).$$

Now that a divides the left hand side, it must divide the right hand side. However, $\gcd(a, b) = 1$. This means that a must divide $y_1 - y$. We can write $y_1 - y = -ak$. Thus, $x - x_1$ must be bk .

Therefore, any solution to $ax + by = c$ has the following form

$$\begin{cases} x = cx_0 + bk, \\ y = cy_0 - ak. \end{cases}$$

Chapter 3 Congruences

3.1 Congruences

Definition 3.1

Assume $n \in \mathbb{N}_2$ and $a, b \in \mathbb{Z}$. We say that a is congruent to b modulo n , denoted by $a \equiv b \pmod{n}$ if and only if n divides $a - b$, i.e., $n|(a - b)$.

Remark Equivalently, $a \equiv b \pmod{n}$ if and only if the remainders are the same when you divide a and b by n .

We have some properties that are easy to check.

Proposition 3.1

Assume $n \in \mathbb{N}_2$, $a, b, c, d \in \mathbb{Z}$, $m \in \mathbb{Z}$, $k \in \mathbb{N}$.

1. $a \equiv a \pmod{n}$.
2. If $a \equiv b \pmod{n}$, then $b \equiv a \pmod{n}$.
3. If $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$ then $a \equiv c \pmod{n}$.
4. If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then $a + b \equiv c + d \pmod{n}$.
5. If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then $ab \equiv cd \pmod{n}$.
6. If $a \equiv b \pmod{n}$, then $a^k \equiv b^k \pmod{n}$.

Lemma 3.1

$a \equiv 0 \pmod{n}$ if and only if $n|a$.

Using Bezout's identity, we can easily show that

Lemma 3.2

Given $a, b \in \mathbb{Z}$ and $n \in \mathbb{N}_2$, the $ax \equiv 1 \pmod{n}$ has a solution in x if and only if $\gcd(a, n) = 1$.

In particular, this tells us that

Lemma 3.3

If p is a prime, a is an integer such that $p \nmid a$, then there exists some a^{-1} such that

$$aa^{-1} \equiv 1 \pmod{p}$$

3.2 Euler ϕ -function

Definition 3.2

Let $n \in \mathbb{N}_1$. We define $\phi(n)$ to be the number of integers between 1 and n that are coprime to n . In other words,

$$\phi(n) = \#\{1 \leq a \leq n : \gcd(a, n) = 1\}$$

Let us first show that for any prime p and $n \in \mathbb{N}_1$, we have

$$\phi(p^n) = p^n - p^{n-1}.$$

We need a lemma which is trivial.

Lemma 3.4

If p is a prime, a is an integer, then $\gcd(a, p^n) = 1$ if and only if $p \nmid a$.

This implies immediately that

$$\phi(p^n) = p^n - p^{n-1}.$$

Moreover, we can show that the Euler ϕ -function is multiplicative.

Lemma 3.5

If m, n are coprime, then $\phi(mn) = \phi(m)\phi(n)$.



This implies, as a corollary, the following.

Lemma 3.6

If $n = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$, then

$$\phi(p_1^{\alpha_1} \cdots p_k^{\alpha_k}) = \phi(p_1^{\alpha_1}) \cdots \phi(p_k^{\alpha_k}) = (p_1^{\alpha_1} - p_1^{\alpha_1-1}) \cdots (p_k^{\alpha_k} - p_k^{\alpha_k-1}).$$



If you wish, we can continue to get the following.

Lemma 3.7

If $n = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$, then

$$\phi(p_1^{\alpha_1} \cdots p_k^{\alpha_k}) = p_1^{\alpha_1} \cdots p_k^{\alpha_k} \left(1 - \frac{1}{p_1}\right) \cdots \left(1 - \frac{1}{p_k}\right) = n \prod_{p|n} \left(1 - \frac{1}{p}\right).$$



In practice, we need to find the solution to $\phi(n) = m$ where m is a fixed positive integer.

Write $n = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$. Then we have

$$\phi(n) = p_1^{\alpha_1-1} \cdots p_k^{\alpha_k-1} \cdot (p_1 - 1) \cdots (p_k - 1).$$

This means that every prime divisor p_i of n must satisfy the condition that $p_i - 1$ divides m . Since each p_i is a prime, and m is a fixed integer, so the possibilities are not too many. We may simplify the problem and continue this process until we get all possible answers.

3.3 Fermat's Little Theorem and Euler's Theorem

Proposition 3.2 (Fermat's Little Theorem)

If p is a prime, a is an integer such that $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}$$



Proposition 3.3 (Euler's Theorem)

If $n \in \mathbb{N}_2$, $a \in \mathbb{Z}$ such that $\gcd(a, n) = 1$, then

$$a^{\phi(n)} \equiv 1 \pmod{n}$$



In particular, we know that $\phi(p) = p - 1$, so that Fermat's Little Theorem is just a special case of Euler's Theorem.

3.4 Wilson's Theorem

Proposition 3.4 (Wilson's Theorem)

If p is an odd prime, then

$$(p-1)! \equiv -1 \pmod{p}.$$



3.5 Chinese Remainder Theorem

Proposition 3.5 (Chinese Remainder Theorem)

If $m_1, \dots, m_n$ are pairwise coprime positive integers, and $a_1, \dots, a_n \in \mathbb{Z}$, then there exists a unique $x_0 \bmod n$ such that for all $1 \leq i \leq n$, we have

$$x_0 \equiv a_i \bmod m_i.$$



I have taught how to compute this x_0 efficiently and correctly in one of my tutorials. Please watch that video if you are not familiar with this sort of computation.